Food Chains & Food Webs

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 25 September 2024

SaveMyExams

Did this video help you? Yes No

Food Chains & Food Webs

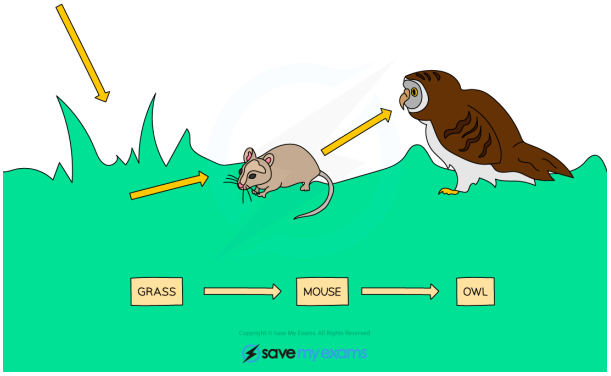
TERM	DEFINITION
PRODUCERS	ORGANISMS THAT PRODUCE **THEIR OWN ORGANIC NUTRIENTS USUALLY USING ENERGY FROM SUNLIGHT**. PLANTS ARE PRODUCERS AS THEY CARRY OUT PHOTOSYNTHESIS TO MAKE GLUCOSE
HERBIVORE	AN ANIMAL THAT GETS ITS ENERGY BY **EATING PLANTS**
CARNIVORE	AN ANIMAL THAT GETS ITS ENERGY BY **EATING OTHER ANIMALS**
PRIMARY CONSUMERS	HERBIVORES – THEY **FEED ON PRODUCERS** (PLANTS)
SECONDARY CONSUMERS	PREDATORS THAT **FEED ON PRIMARY CONSUMERS**
TERTIARY CONSUMERS	PREDATORS THAT **FEED ON SECONDARY CONSUMERS**
DECOMPOSERS	BACTERIA AND FUNGI THAT GET THEIR ENERGY FROM **FEEDING OFF DEAD AND DECAYING ORGANISMS AND UNDIGESTED WASTE** (SUCH AS FAECES) BY SECRETING ENZYMES TO BREAK THEM DOWN

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



A food chain with three trophic levels

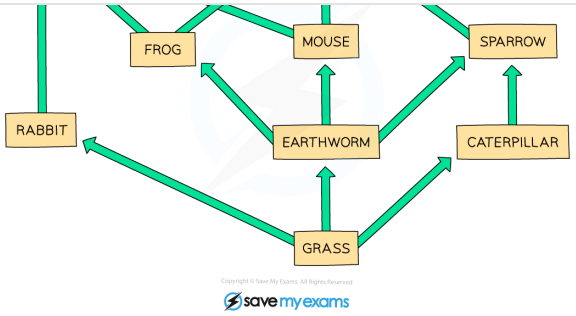
- A food chain shows the transfer of energy from one organism to the next, starting with a producer
- The source of all energy in a food chain is **light energy from the Sun**
- The arrows in a food chain show the **transfer of energy** from one trophic level to the next
- Energy is transferred from one organism to another by ingestion (eating)
- In the food chain above:

Food chain example table

Position in food chain	Organism	Explanation
Producer	Grass	Makes its own food using energy from sunlight in photosynthesis
Primary consumer	Mouse	Eats the producer
Secondary consumer	Barn owl	Eats the primary consumer

Food Webs

- A **food web** is a network of interconnected food chains
- Food webs are more realistic ways of showing connections between organisms within an ecosystem as **animals rarely exist on just one type of food source**



A food web shows the interdependence of organisms

- Food webs give us a lot more information about the transfer of energy in an ecosystem
- They also show **interdependence** - how the change in one population can affect others within the food web
- For example, in the food web above, if the **population of earthworms decreased**:
  - The population of **grass plants would increase** as there are now fewer species feeding off them
  - The populations of **frogs and mice would decrease significantly** as earthworms are their only food source
  - The population of **sparrows would decrease slightly** as they eat earthworms but also have another food source to rely on (caterpillars)

- Most of the changes in populations of animals and plants happen as a result of **human impact** - either by **overharvesting of food species** or by the **introduction of foreign species to a habitat**
- Due to interdependence, these can have **long-lasting knock-on effects** to organisms throughout a food chain or web

Examiner Tips and Tricks

Questions about interdependence in food webs are common and easy to gain marks on if you answer them fully and correctly.

Do not say an animal or plant would 'die out' as this is unlikely to happen – stick to using the words decrease or increase. If in doubt, always give your reason for the increase or decrease in population.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.